

Structure of Atom

CHAPTER 9

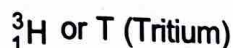
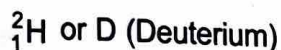
THIS CHAPTER INCLUDES

ISOTOPES

Isotopes are the atoms of same element which have same atomic number (number of protons) but different mass numbers.

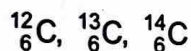
e.g.,

(i) Isotopes of hydrogen

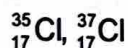


Note : Ordinary hydrogen (${}^1_1\text{H}$) is the only atom that has no neutron.

(ii) Isotopes of carbon



(iii) Isotopes of chlorine



The chemical properties of isotopes are similar but their physical properties are different.

Average Atomic Mass of Isotopes

$$\text{Average atomic mass} = \frac{\sum_{i=1}^n \% \text{abundance} \times \text{Atomic mass of isotopes}}{100}$$

Example 1 : The average atomic mass of a sample of an element Y is 16.2 u. Calculate the percentages of isotopes ${}^{18}_8\text{Y}$ and ${}^{16}_8\text{Y}$ in this sample.

Solution : Let the percentage of isotope of ${}^{18}_8\text{Y} = 'x'$ and the percentage of isotope of ${}^{16}_8\text{Y} = (100 - x)$.

On the basis of above data, percentage will be

$$\Rightarrow \left[\left(18 \times \frac{x}{100} \right) + 16 \times \frac{(100 - x)}{100} \right] = 16.2$$

$$\Rightarrow \left[\frac{18x + 1600 - 16x}{100} \right] = 16.2$$

$$\Rightarrow \left[\frac{2x + 1600}{100} \right] = 16.2$$

$$\Rightarrow 2x + 1600 = 1620$$

$$\Rightarrow 2x = 1620 - 1600$$

$$\Rightarrow 2x = 20$$

$$\Rightarrow x = 10$$

Hence, percentage of $^{18}_8\text{Y}$ is 10% and percentage of $^{16}_8\text{Y}$ is 90%.

Example 2 : The percentage abundance of two isotopes of bromine $^{79}_{35}\text{Br}$ and $^{81}_{35}\text{Br}$ are 49.7% and 50.3% respectively. Calculate average atomic mass of bromine atom.

Solution : On the basis of above data, average atomic mass of bromine atom will be

$$\Rightarrow \left(\frac{79 \times 49.7}{100} \right) + \left(\frac{81 \times 50.3}{100} \right)$$

$$= \left(\frac{3926.3}{100} + \frac{4074.3}{100} \right)$$

$$= \left(\frac{8000.6}{100} \right)$$

$$= 80.006 \text{ u}$$

Hence, average atomic mass will be 80.006 u.

Uses of isotopes

- (i) An isotope of uranium (uranium-235) is used as a fuel in nuclear reactors.
- (ii) An isotope of cobalt (cobalt-60) is used in the treatment of cancer.
- (iii) An isotope of iodine (iodine-131) is used in the treatment of goitre.
- (iv) An isotope of phosphorus (P-32) is used in the treatment of leukemia.
- (v) Arsenic-74 is employed to detect tumours.
- (vi) Blood clots in the circulatory system are located by sodium-24.

Isobars

Isobars are the atoms of different elements with different atomic numbers but same mass number.

e.g., Argon $^{40}_{18}\text{Ar}$ and calcium $^{40}_{20}\text{Ca}$ are isobars of each other.

In isobars, sum of neutron and proton ($n + p$) is same but they have different number of protons.

Isotones

Isotones are the atoms of different elements with same number of neutrons.

e.g., $^{13}_6\text{C}$ and $^{14}_7\text{N}$ are isotones of each other as –

$$^{13}_6\text{C} : \text{neutrons} = 13 - 6 = 7$$

$$^{14}_7\text{N} : \text{neutrons} = 14 - 7 = 7$$

Isoelectronic Species

Those species (atoms, ions or molecules) that have same number of electrons are called as isoelectronic species.

e.g.,

$$^1_1\text{H} : \text{one } e^- ; \quad \text{Na}^+ : 10 e^- ; \quad \text{Al}^{3+} : 10 e^-$$

$$\text{He}^+ : \text{one } e^- ; \quad \text{Ne} : 10 e^- ; \quad \text{F}^- : 10 e^-$$

H and He^+ are isoelectronic species and also, Na^+ , Ne, Al^{3+} , F^- are isoelectronic species.

Isodiaphers

The atoms of different elements that show same $(n - p)$ value, are called as Isodiaphers of each other.

e.g.,

$$\text{Uranium} (^{238}_{92}\text{U}) ; \quad \text{Thorium} (^{234}_{90}\text{Th})$$

$$(n) \text{ Neutrons} : 238 - 92 = 146 ; \quad 234 - 90 = 144$$

$$(p) \text{ Protons} : 92 ; \quad 90$$

$$(n - p) : 146 - 92 = 54 ; \quad 144 - 90 = 54$$

$\therefore$ $^{238}_{92}\text{U}$ and $^{234}_{90}\text{Th}$ are isodiaphers of each other.

$^{39}_{19}\text{K}$ and $^{19}_9\text{F}$ are isodiaphers of each other.

Isosteres

These are molecules or ions with the same number of atoms and the same number of valence electrons.





Assignment

Single answer type questions:

- Which of the following group represents isoelectronic species?
 - Cl^- , Ar, K^+
 - Ne, F^+ , O^{2-}
 - N^{3-} , O^{2-} , Ar
 - Ar, Na^+ , Mg^{2+}
- Which of the following pairs are isosteres?
 - CO_2 , NO_2
 - CO_2 , N_2O
 - H_2O , OF_2
 - CO_2 , H_2S
- $^{13}_6\text{C}$ and $^{14}_6\text{C}$ are
 - Isotopes
 - Isobars
 - Isotones
 - Isodiaphers
- Which of the following pairs represent an isobar?
 - 12 P + 13 N and 15 P + 13 N
 - 12 P + 14 N and 13 P + 13 N
 - 13 P + 13 N and 10 P + 13 N
 - 13 P + 14 N and 12 P + 14 N
- The triad of nuclei that is isotonic is
 - $^{14}_6\text{C}$, $^{15}_7\text{N}$, $^{17}_9\text{F}$
 - $^{12}_6\text{C}$, $^{14}_7\text{N}$, $^{19}_9\text{F}$
 - $^{14}_6\text{C}$, $^{14}_7\text{N}$, $^{17}_9\text{F}$
 - $^{14}_6\text{C}$, $^{14}_7\text{N}$, $^{19}_9\text{F}$
- Which of the following have same chemical properties but have different physical properties?
 - Isotones
 - Isobars
 - Isoelectronic species
 - Isotopes
- Which one of the following sets of ions represents a collection of isoelectronic species?
 - N^{3-} , O^{2-} , F^- , S^{2-}
 - Li^+ , Na^+ , Mg^+ , Ca^{2+}
 - K^+ , Cl^- , Ca^{2+} , Sc^{3+}
 - Ba^{2+} , Sr^{2+} , K^+ , Ca^{2+}
- If velocity of an electron in 1st orbit of H atom is V , what will be the velocity of 3rd orbit of Li^{+2} ?
 - V
 - $V/3$
 - $3V$
 - $9V$
- The orbital with zero angular momentum is
 - s
 - p
 - d
 - f
- Which of the following is electronic configuration of Cu^{2+} ($Z = 29$)?
 - $[\text{Ar}]4s^1 3d^8$
 - $[\text{Ar}]4s^2 3d^{10} 4p^1$
 - $[\text{Ar}]4s^1 3d^{10}$
 - $[\text{Ar}] 3d^9$
- Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon?
 - 3s
 - 2p
 - 2s
 - 1s
- The total number of neutrons in dipositive zinc ions with mass number 70 is
 - 34
 - 40
 - 36
 - 38
- The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$. This represents its
 - Excited state
 - Ground state
 - Cationic form
 - None
- If radius of second stationary orbit (in Bohr's atom) is R , then radius of third orbit will be
 - $R/3$
 - $9R$
 - $R/9$
 - $2.25 R$
- Which orbital is non-directional?
 - s
 - p
 - d
 - All of these

16. Which orbit would be the first to have 'g' subshell?
 (1) 3^{rd} (2) 4^{th}
 (3) 5^{th} (4) 6^{th}
17. The decreasing order of energy of the 3d, 4s, 3p, 3s orbitals is
 (1) $3d > 3s > 4s > 3p$ (2) $3s > 4s > 3p > 3d$
 (3) $3d > 4s > 3p > 3s$ (4) $4s > 3d > 3s > 3p$
18. The number of d-electrons retained in Fe^{2+} (At. no. of Fe = 26) ion is
 (1) 6 (2) 3
 (3) 4 (4) 5
19. Which one of the following sets of ions represents the collection of isoelectronic species?
 (1) $\text{Na}^+, \text{Mg}^{2+}, \text{Al}^{3+}, \text{Cl}^-$ (2) $\text{Na}^+, \text{Ca}^{2+}, \text{Sc}^{3+}, \text{F}^-$
 (3) $\text{K}^+, \text{Cl}^-, \text{Mg}^{2+}, \text{Sc}^{3+}$ (4) $\text{K}^+, \text{Ca}^{2+}, \text{Sc}^{3+}, \text{Cl}^-$
20. According to Bohr's theory angular momentum of electron in 5^{th} shell is
 (1) $1.0 \text{ h}/\pi$ (2) $10 \text{ h}/\pi$
 (3) $2.5 \text{ h}/\pi$ (4) $25 \text{ h}/\pi$
21. The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom?
 (1) $\text{He}^+ (n = 2)$ (2) $\text{Li}^{2+} (n = 2)$
 (3) $\text{Li}^{2+} (n = 3)$ (4) $\text{Be}^{3+} (n = 2)$
22. If r is radius of first orbit of He^+ , the radius of n^{th} orbit of He^+ will be
 (1) $\frac{2r}{n^2}$ (2) $\frac{r}{2n^2}$
 (3) n^2r (4) $\frac{r}{n^2}$
23. An electron is moving in 3^{rd} orbit of Li^{+2} . The radius is
 (1) $1.59 \text{ \AA}$ (2) $0.529 \text{ \AA}$
 (3) $0.176 \text{ \AA}$ (4) $2.587 \text{ \AA}$
24. Maximum number of electrons in an orbit can be
 (1) n^2
 (2) $(2n + 2)$
 (3) $(2n + 1)$
 (4) $2n^2$
25. Ratio of angular momentum of 3^{rd} and 1^{st} orbit is
 (1) $\sqrt{3} : 1$ (2) $\sqrt{6} : \sqrt{2}$
 (3) $3 : 1$ (4) $9 : 1$
26. Which of the following relation is correct according to Bohr's model?
 (1) $\text{PE} = -\text{KE}$ (2) $\text{Velocity} \propto n$
 (3) $r_n \propto \frac{n^2}{Z}$ (4) All of these
27. Which hydrogen like species will have same radius as that of first Bohr orbit of a hydrogen atom?
 (1) $n = 2, \text{Li}^+$
 (2) $n = 2, \text{Be}^{+3}$
 (3) $n = 2, \text{He}^+$
 (4) $n = 3, \text{Li}^{+2}$
28. The orbital angular momentum of an electron in 2s and 2p orbital respectively
 (1) $\frac{1}{2}, \frac{h}{2\pi}$ (2) Zero, $\sqrt{2} \frac{h}{2\pi}$
 (3) $\frac{h}{2\pi}$, Zero (4) $\sqrt{2} \frac{h}{2\pi}, \sqrt{6} \frac{h}{2\pi}$
29. Which of the following element have five unpaired electron in d-subshell?
 (1) Iron (2) Chromium
 (3) Scandium (4) Zinc
30. Total number of shells present in hydrogen atom is/are
 (1) One (2) Two
 (3) Three (4) Infinite
31. Which shell would be the first to have 'f' subshell?
 (1) K (2) M
 (3) N (4) O
32. Which of the following ion has maximum number of unpaired d-electrons?
 (1) Zn^{+2} (2) Fe^{+3}
 (3) Cr^{+3} (4) Mn^{+4}

33. The possible values of magnetic quantum number for p -orbital are
- (1) 0
 - (2) -1, 0, +1
 - (3) -2, -1, 0, +1, +2
 - (4) -3, -2, -1, 0, +1, +2, +3
34. In multi-electron atom $4s$ -orbital is lower in energy than
- (1) $3d$ -orbital
 - (2) $3p$ -orbital
 - (3) $2s$ -orbital
 - (4) $2p$ -orbital
35. Shape of an orbital is given by
- (1) Principal quantum number
 - (2) Spin quantum number
 - (3) Azimuthal quantum number
 - (4) Magnetic quantum number
- More than one answer type questions:**
36. The sum of the number of neutrons and protons in the isotopes of hydrogen is
- (1) 1
 - (2) 2
 - (3) 4
 - (4) 3
37. Bohr's theory is not applicable to
- (1) He
 - (2) Li^{2+}
 - (3) He^{2+}
 - (4) The H-atom
38. Choose the correct relation on the basis of Bohr's theory
- (1) Velocity of electron $\propto \frac{1}{n}$
 - (2) Frequency of revolution $\propto \frac{Z^2}{n^3}$
 - (3) Radius of orbit $\propto n^2 Z$
 - (4) Force on electron $\propto \frac{Z^3}{n^4}$
39. According to Bohr's model, which statement is correct?
- (1) Every orbit has fixed energy
 - (2) Angular momentum is quantised
 - (3) $TE = -KE$
 - (4) Energy of an electron in the orbit changes with time
40. For $n = 4$, which one of the following values of l is not possible?
- (1) -1
 - (2) 2
 - (3) 3
 - (4) 4

